LEARNING ROADMAP · 2024

30-DAY AI ENGINEER

Transition from Data Professional · 2 Hours Daily · 60 Hours Total

30	4	2H	60H
DAYS	WEEKS	DAILY	TOTAL

✦ WEEKLY BREAKDOWN

WEEK 01 · DAYS 1–7

AI FUNDAMENTALS & DEEP LEARNING BASICS

Foundation

Understand AI/ML pipeline & build first neural network

DAYS 1–3 · AI ENGINEERING OVERVIEW

- Watch: "AI Engineering vs Data Science" (YouTube)
- Read: AI system architecture basics
- Practice: Review Python OOP concepts

DAYS 4–7 · NEURAL NETWORKS & TENSORFLOW

- 30 min: TensorFlow basics tutorials
- 60 min: Code along neural network examples
- 30 min: Document learnings in Jupyter notebooks

RESOURCES

Fast.ai Practical Deep Learning Course (Part 1) · "Hands-On Machine Learning" by Aurélien Géron (Ch. 10–11) · Google's Machine Learning Crash Course

WEEKEND PROJECT

Build a digit classifier using MNIST dataset

WEEK 02 · DAYS 8–14

DEEP LEARNING FRAMEWORKS & COMPUTER VISION

Core AI

Deploy first computer vision model

DAYS 8–11 · PYTORCH & COMPUTER VISION

- 45 min: PyTorch fundamentals
- 45 min: CNN architecture study
- 30 min: Image preprocessing techniques

DAYS 12–14 · MODEL TRAINING & OPTIMIZATION

- 60 min: Transfer learning implementation
- 30 min: Hyperparameter tuning
- 30 min: Model evaluation metrics

RESOURCES

PyTorch tutorials official website · CS231n Stanford lectures (selected videos) · Kaggle Learn Computer Vision

WEEKEND PROJECT

Create an image classification API using FastAPI

WEEK 03 · DAYS 15–21

NLP & LARGE LANGUAGE MODELS

LLM Era

🎯 Build a text generation application

DAYS 15–18 · NLP FUNDAMENTALS & TRANSFORMERS

- 45 min: NLP preprocessing (tokenization, embeddings)
- 45 min: Transformer architecture study
- 30 min: Hugging Face tutorials

DAYS 19–21 · WORKING WITH PRE-TRAINED MODELS

- 60 min: Fine-tuning BERT/GPT models
- 30 min: Prompt engineering basics
- 30 min: LangChain introduction

RESOURCES

Hugging Face NLP Course · "Natural Language Processing with Transformers" book · Andrew Ng's ChatGPT Prompt Engineering course

WEEKEND PROJECT

Build a chatbot using OpenAI API or open-source LLM

WEEK 04 · DAYS 22–30

MLOPS & PRODUCTION DEPLOYMENT

Ship It

🎯 Deploy end-to-end AI application

DAYS 22–25 · MLOPS ESSENTIALS

- 45 min: Docker containerization for ML
- 45 min: Model versioning with MLflow
- 30 min: CI/CD for ML pipelines

DAYS 26–28 · CLOUD DEPLOYMENT & MONITORING

- 60 min: Deploy models on AWS/GCP/Azure
- 30 min: Model monitoring setup
- 30 min: Performance optimization

DAYS 29–30 · INTEGRATION & PORTFOLIO

RESOURCES

"Building Machine Learning Powered Applications" · MLOps Specialization on Coursera · Made With ML MLOps course

FINAL CAPSTONE PROJECT

› Data pipeline (using existing SQL/Python skills)
› Model training & evaluation
› API deployment
› Simple web interface
› Monitoring dashboard

✦ FINAL OUTCOME

🎉 YOU ARE NOW AN AI ENGINEER

Consistency beats intensity — 2 hours daily for 30 days = 60 hours of focused learning

- ✅ Built 5+ AI projects (CV, NLP, deployment)
- ✅ Hands-on: TensorFlow, PyTorch, Transformers
- ✅ Deployed models to production
- ✅ Portfolio showcasing end-to-end AI solutions
- ✅ Strong foundation for AI engineering roles

✦ NEXT STEPS

CONTRIBUTE
Open-source AI projects

SPECIALISE
Computer Vision or NLP

ADVANCED
Reinforcement Learning, GANs

BUILD
Domain-specific AI solutions

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